

SAFETY DATA SHEET

Section 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier	Caprolactam (epsilon-Caprolactam) 6-Hexanelactam
1.2 Relevant identified uses of the substance or mixture and uses advised against	Raw material of Polyamide6 (Nylon6)
1.3 Details of the supplier of the safety data sheet	UBE Chemicals (Asia) Public Company Limited Address: 18 th Floor, Sathorn Square Office Tower, 98 North Sathorn Road, Silom, Bangrak, Bangkok 10500, Thailand TEL: +66-2-206-9300 FAX: +66-2-206-9316 E-mail: sds.ube-cpl@ube.co.th
1.4 Emergency telephone number	UBE Chemicals (Asia) Public Company Limited TEL: +66-2-206-9300 (within business hours)

Section 2: Hazards identification

2.1 Classification of the substance or mixture

GHS Classification	Acute toxicity - inhalation, Category 4, H332 Acute toxicity - oral, Category 4, H302 Skin corrosion/irritation, Category 2, H315 Eye damage/irritation, Category 2, H319 Specific target organ toxicity - single exposure, Category 3 (Respiratory tract irritation), H335
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2.2 Label elements



Signal word	Warning
Hazard statements	H332: Harmful if inhaled H302: Harmful if swallowed H315: Causes skin irritation. H319: Causes serious eye irritation. H335: May cause respiratory irritation.
Precautionary statements	
prevention	P271: Use only outdoors or in a well-ventilated area. P280: Wear protective gloves/eye protection/face protection. P280: Wear eye protection/face protection.
response	P312: Call a POISON CENTER or doctor/physician if you feel unwell. P305 + P351 + P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

	P304+P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
storage	None
disposal	None
Supplemental information	None.
2.3 Other hazards	None.
2.4 The main symptoms and emergency summary after exposure	Not found

Section 3: Composition/information on ingredients

3.1 Substances

<i>Declarable components</i>	<i>Conc. (wt%)</i>	<i>EC No.</i>	<i>CAS No.</i>
Caprolactam	Min. 99.9	203-313-2	105-60-2

Section 4: First aid measures

4.1 Description of first aid measures

Inhalation	Remove victim to fresh air. Get medical attention.
Skin	Wash away with soapsuds or plenty of water.
Eye	Immediately flush with copious amount of water for at least 15 minutes (remove contact lenses if easily possible), and get medical attention.
Ingestion	Rinse mouth. Do NOT induce vomiting. Refer for medical attention.

4.2 Most important symptoms and effects, both acute and delayed

Not available.

4.3 Indication of any immediate medical attention and special treatment needed

Not available.

Section 5: Firefighting measures

5.1 Extinguishing media

Suitable	Spray of water, Carbon dioxide or Powder extinguisher
Unsuitable	None.

5.2 Special hazards arising from the substance or mixture

In case of combustion, toxic fumes (carbon dioxide, carbon monoxide, nitrogen oxides, hydrocyanic acid, etc.) are emitted.

5.3 Advice for firefighters

Remove containers from fire or cool them with water spray. Firefighters should wear an approved self-contained breathing apparatus and full protective clothing.

5.4 Special protective

Put protective equipment such as self-contained breathing apparatus

device to protect the fire fighters. and fire fighter suit etc.

Section 6: Accidental release measures

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| 6.1 Personal precautions, protective equipment and emergency procedures | The substance irritates the skin. The vapour irritates the eyes, the skin and the respiratory tract. So should treat the substance with a powder protection mask, safety goggles and rubber gloves. |
| 6.2 Environmental precautions | This substance may be hazardous to the environment. Do not eject the substance directly into the river. |
| 6.3 Methods and material for containment and cleaning up | Gather and collect. In the case, work as carefully as possible not to raise dust.
When it is liquid, gather after cooling and solidifying. |
| 6.4 Reference to other sections | For recommended personal protective equipment, see Section 8.
For disposal considerations, see Section 13. |
| 6.5 Precautionary measures to prevent secondary hazards | Wear protective equipment and prevent spilled material from making contact with eyes, skin and clothing. |

Section 7: Handling and storage

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| 7.1 Precautions for safe handling | Always need to wear a powder protection mask, safety goggles and rubber gloves to avoid inhalation and eyes and skin-contact. |
| 7.2 Conditions for safe storage, including any incompatibilities | Because it has water absorption property, put in an airtight container. Avoid direct sunlight, keep in a cool, dark and ventilated place. |
| 7.3 Specific end use(s) | In industrial situations, hot molten material may cause thermal burns. |

Section 8: Exposure controls/personal protection

8.1 Control parameters

EU limit values	None.
US limit values	ACGIH(2014) 5 mg/m ³ (TWA, Inhalable fraction and vapor)
Other: human health (DNELs, DMELs)	Not available
Other: environmental (PNEC)	PNEC aqua (freshwater): 2 mg/L (Assessment factor 50) PNEC sediment (freshwater): 18.7 mg/kg sediment dw PNEC soil: 2.55 mg/kg soil dw

8.2 Exposure controls

Engineering controls	In case of using flake caprolactam in doors, install the local ventilation equipment. Install the hand wash station and the eyewash station near the workplace.
Personal protective equipment	Respiratory protection: Powder protection mask Hand protection: Rubber gloves Chemical resistant protective gloves (EN 374) Suitable materials also with prolonged, direct contact (Recommended: Protective index 6,

corresponding > 480 minutes of permeation time according to EN 374):

Butyl rubber, BR, Recommended thickness l: ≥ 0.7 mm

Nitrile rubber, NBR, Recommended thickness: ≥ 0.4 mm

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer.

- Penetration time of glove material

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

Eye protection: Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Skin protection: Clothing to prevent skin exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

Environmental exposure controls Do not eject it directly into the river.

Section 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance	Flake: White powder easily deliquescent Molten (Liquid): Transparent
Odour	Faint odor
Odour threshold	Not available
pH	7 – 8.5
Melting/freezing point	69.3°C
Initial boiling point/range	270.8°C (1013.25hPa)
Flash point	125°C
Evaporation rate	Not available
Flammability (solid, gas)	Not available
Flamm. or expl. limits	Not available
Vapour pressure	0.13Pa (20°C)
Vapour density	3.91 (air=1)
Relative density	1.105 (20°C), 1.015 (78.5°C)
Solubilities	Water solubility: 866.89g/L (22°C) Fat solubility: Easily soluble to ethyl alcohol, ether, chloroform, benzene.
Partition coeff. (K_{ow})	Log Pow: 0.12 (25°C)
Auto-ignition temp.	395°C (1013mbar)
Decomposition temp.	Not available
Viscosity	Not available
Explosive properties	1.4 to 8.0% by volume
Oxidising properties	No oxidising properties

9.2 Other information

Not available

Section 10: Stability and reactivity

10.1 Chemical stability	Stable under normal temperature (25°C) and pressure (1atm).
10.2 Possibility of hazardous reactions	Not available
10.3 Conditions to avoid	Exposure to ignition source (heat, flame, spark, etc.), moisture, and oxidizing reagents.
10.4 Incompatible materials	Strong oxidizing reagents
10.5 Hazardous decomposition products	Emit toxic fumes (nitrogen oxides, ammonia, etc.) by exposure to ignition source or strong oxidizing reagents.

Section 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity	LD ₅₀ = 1475 mg/kg (Oral, rat) LC ₅₀ = 8.16 mg/L, 4 hours (Inhalation, rat) LD ₅₀ > 2000 mg/kg (Dermal, rat)
Skin corrosion/irritation	Mild irritating Mild signs of irritation were observed in a skin sensitization test in which 0.4ml of a 75% CAP-solution to guinea pigs (Springborn, 1991).
Serious eye damage/irritation	Moderate irritating - Eyes of rabbits Eyelid closure and eye discharge was observed for irritation by inhalation.
Respiratory or skin sensitisation	Respiratory tract : Not sensitizing in Alarie assay. Skin: Not sensitizing in guinea pig maximization test and Buehler test.
Germ cell mutagenicity	in vitro Bacterial Reverse Mutation Test (Ames test, OECD 471): Negative Mammalian cell gene mutation assay: Negative Bacterial gene mutation assay (gene mutation): Negative In vivo mammalian chromosome aberration test: Negative (OECD 473) Gene mutation test: Ambiguous Chromosome aberration assay: Negative (OECD 475) DNA damage and/or repair): Negative
Carcinogenicity	No indication for a carcinogenic potential was identified in carcinogenicity studies in rodents. ACGIH: A5 (Not suspected as a human carcinogen) IARC Monographs: Group 4 (Probably not carcinogenic to humans)
Reproductive toxicity	Caprolactam was not teratogenic or embryotoxic in rats and rabbits when given orally at high doses. (NOAEL _{tera} 250 mg/kg bw/day).
STOT-single exposure	Following clinical signs of toxicity were observed for irritation : eyelid closure, irregular respiration, escape reaction during exposure, dyspnoea, piloerection, bloody nose, eye discharge, tremor after exposure.
STOT-repeated exposure	Only local respiratory irritation was observed on inhalation route of exposure. No systemic effects was observed.
Aspiration hazard	Not available.

Section 12: Ecological information

12.1 Toxicity	Fish(<i>Oryzias latipes</i>): LC ₅₀ >100 mg/L, 96hours Water flea(<i>Daphnia magna</i>) EC ₅₀ > 1000mg/L, 48hours Algae: EC ₅₀ > 1000mg/L, 72hours
12.2 Persistence and degradability	Readily biodegradable 82% after 14 days (O ₂ consumption, OECD 301 C) Degradation rate in water: 4.7*10 ⁻² [d ⁻¹] Degradation rate in air: 0.795 [d ⁻¹]
12.3 Bioaccumulative potential	Due to the low log Pow(<3), accumulation in organisms is not expected.
12.4 Mobility in soil	Due to the log Koc(1.76 at 20°C), adsorption of the substance to the solid soil phase (e. g. clay) is not expected.
12.5 Other adverse effects	Not available.

Section 13: Disposal considerations

13.1 Waste treatment methods	After solving into solvents which easily can be burn, or after making it mixture burn in an incinerator where an afterburner and a scrubber are set. Treat with activated sludge process or ask authorized industrial waste treatment agents which have capability of treatment. Comply with all federal, state and local regulations. Do not dump this material into sewers, on the ground or into any body of water.
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Section 14: Transport information

14.1 UN Number	No classification assigned
14.2 UN proper shipping name	None
14.3 Transport hazard class(es)	None
14.4 Packing group	None
14.5 Environmental hazards	Not classified as environmentally hazardous.
14.6 Special precautions for user	Not available.
14.7 Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code	Product name: epsilon-Caprolactam (molten or aqueous solutions) Pollution category: Z

Section 15: Regulatory information

15.1 Safety, health and environmental	Please refer to any other regulations of each country.
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regulations/legislation
specific for the
substance or mixture

Section 16: Other information

Revisions Section 1.3 Change address

References

Training advice Not available.

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